

Recording Activity: Manual Voltage Mapping

Name: _____ Date: _____

Student: Write predictions and record measurements only after the lesson’s teacher checkpoint. **Teacher:** Verify the wiring and meter setup before USB power is connected. This worksheet does not replace the safety and wiring directions in the lesson.

Predict

What do you predict will happen to the measured voltage as the control moves from one end to the other?

Measure and Map

Record the minimum, maximum, and at least six additional readings. Use the lesson lookup table to fill in the last two columns. If eight readings do not include every decimal level, use the additional rows and keep measuring until they do.

Control position	Measured voltage	Decimal level (0-7)	3-bit value
Minimum			
Reading 2			
Reading 3			
Reading 4			
Reading 5			
Reading 6			
Reading 7			
Maximum			
Additional reading 1			
Additional reading 2			
Additional reading 3			
Additional reading 4			

Check each decimal level after it appears in your completed table:

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

Compare and Explain

- Find two nearby voltage readings that have the same decimal level.
First voltage: _____ V Second voltage: _____ V
Shared decimal level: _____ Shared 3-bit value: _____
Why can both readings share one level?

- How many different decimal levels are in this activity? How many 3-bit values?

- In your own words, explain how the lookup table turns a smoothly changing voltage into digital choices.
